End Semester exam
EE2025 - Engineering Electromagnetics
November 23, 2023
50 marks
Exam Time: 9:00 am - 12:00 pm

Note:

- Submit the question paper, formula sheet, along with the answer paper.
- Answers must have a clear explanation and relevant steps. Don't forget to include units!

Part A This part has 10 questions with 2 marks each.

1. A 50 Ω lossless line terminated in a purely resistive load has a voltage standing-wave ratio of 3. Find all possible values of Z_L.

Solution:

$$|\Gamma_L| = \frac{\rho - 1}{\rho + 1} = \frac{3 - 1}{3 + 1} = 0.5$$

For a purely resistive load, $\phi_L = 0$ or π. For $\phi_L = 0$, $\Gamma_L = 0.5$

$$Z_L = Z_0 \frac{1 + \Gamma_L}{1 - \Gamma_L} = 50 \frac{1 + 0.5}{1 - 0.5} = 150\Omega$$

For $\phi_L = \pi$, $\Gamma_L = -0.5$ and

$$Z_L = Z_0 \frac{1 + \Gamma_L}{1 - \Gamma_L} = 50 \frac{1 - 0.5}{1 + 0.5} = (50/3)\Omega$$

Marking scheme: One mark for Z_L when ϕ_L is zero and one mark for Z_L when ϕ_L is π. No units no mark.

2. A quarter wave transformer (QWT) is inserted between a transmission line of characteristic impedance Z_0 and a real load with impedance R? What should be the magnitude of the QWT impedance? Does its presence mean no reflection happens at the QWT - load interface?

Solution:

The QWT impedance should be $\sqrt{RZ_0}$. Reflection does happen at that interface but the length of the QWT ensures that there is a destructive interference of the incident and reflected signals, so overall there is no reflected wave.

Marking scheme: One mark for the expression $\sqrt{RZ_0}$ and one mark for answer with explanation for the second part.

3. Define the plane of incidence for an electromagnetic wave travelling in a homogeneous medium and incident on a dielectric interface. Sketch the electric and magnetic field directions and the direction of propagation of an incident TE wave with respect to the plane of incidence.

Solution: The plane if incidence is the plane that contains the propagation or k vector of the incident wave ($\hat{k_i}$), the reflected wave ($\hat{k_r}$) and the normal to the surface ($\hat{n}$). For a TE wave, the electric field is perpendicular to the plane of incidence. Therefore, the magnetic field lies in the plane of incidence, as shown in Fig. 1.

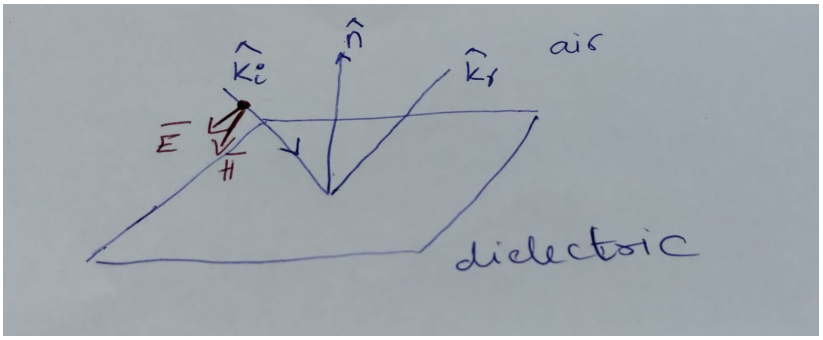


Figure 1: Sketch of Wave vector

Marking scheme:

Definition of plane of incidence **1 mark** : Contains Incident vector, reflected vector, transmitted vector and normal of the dielectric interface. If any two vectors are mentioned marks are awarded.

Sketch of incident wave with electric and magnetic field - **1 marks**

4. The wavenumber and phase velocity of a 5 GHz plane wave travelling in a medium with permittivity and permeability values of $\epsilon = 3\epsilon_0$ and $\mu = 3\mu_0$ are

   A. $100\pi\ rad.m^{-1}$ and 3×10^8 m/s

   B. $300\pi\ rad.m^{-1}$ and 10^8 m/s

   C. $100\pi\ rad.m^{-1}$ and 10^8 m/s

   D. None of the above

Solution: Option (c). Phase velocity (v_p),

$$v_p = \frac{1}{\sqrt{\mu\epsilon}} = \frac{3 \times 10^8}{3} = 10^8 m/s$$

Wavenumber (k)

$$k = \omega\sqrt{\mu\epsilon} = \frac{2\pi \times 5 \times 10^9}{10^8} = 100\pi \quad rad.m^{-1}$$

5. Figure 2 shows two configurations of metal plates. In both cases, the plates are of opposite charge. Plot the electric field lines between the plates.

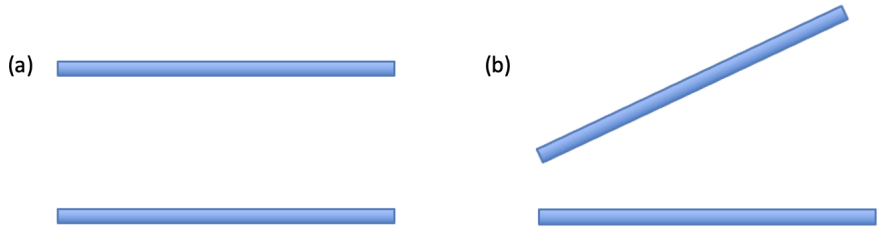


Figure 2: Figure for Qn. 7

Solution: Two key features need to be shown in the drawing. First, the electric field is normal to the metal plates. Second, the density of lines is higher when the spacing is lower. This is because the plates are equipotential surfaces and the potential difference between them is given by the line integral $\int \boldsymbol{E} \cdot \boldsymbol{dl}$ where the path can be one of the field lines shown in the figure below. Therefore, for a fixed value of the line integral, the strength of the electric field will need to be higher where the plates are closer.

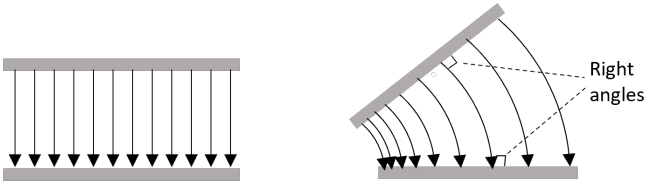


Figure 3: Field lines between metal plates for cases (a) and (b) respectively.

Marking scheme

- 1 mark for figure (a). Pointing the right angles and with explanation
- 1 mark for figure (b). Pointing out that the field lines are curved due the variation in distance between the plates. Also the lines should be denser from left to right indicating string electric field when the plates are closer.

6. You are given the Poynting vector on the surface of a long straight conducting wire, carrying dc current I, of radius b and conductivity σ as

$$\boldsymbol{P} = -\hat{a_r}\frac{I^2}{2\sigma\pi^2 b^3}$$

What equation do you get if you use Poynting's theorem, that confirms this is correct?

Solution: **Derivation (1 mark)** Poynting theorem states,

$$\oint_S (\boldsymbol{E} \times \boldsymbol{H}) \cdot \boldsymbol{da} = -\frac{\partial}{\partial t}(\oint_V \frac{\mu}{2}|\boldsymbol{H}|^2 dv + \oint_V \frac{\epsilon}{2}|\boldsymbol{E}|^2 dv) - \oint_V \sigma|\boldsymbol{E}|^2 dv$$

Let us consider a cylindrical volume with radius 'b' and height 'h' whose axis is along the axis of the wire. Since the fields are static the time derivative term is zero. The electric field in the volume is everywhere uniform and equal to $\boldsymbol{J}/\sigma = \boldsymbol{I}/(\pi b^2\sigma)$.Therefore,

$$\oint_S \boldsymbol{P}\cdot\boldsymbol{da} = -\sigma\frac{I^2}{\pi^2 b^4\sigma^2}(\pi b^2 h) = \frac{I^2 h}{\pi b^2\sigma}$$

The poynting vector is radially inward everywhere and the magnitude is only dependent on the radial distance from the axis of the wire. The area vector $\boldsymbol{da}$ for the circular faces, of the volume under consideration, are perpendicular to the poyting vector and do not contribute to the integral. Thus,

$$|\boldsymbol{P}|(2\pi bh) = \frac{I^2 h}{\pi b^2\sigma} \implies |\boldsymbol{P}| = \frac{I^2}{2\sigma\pi^2 b^3}$$

So $\boldsymbol{P} = \frac{I^2}{2\sigma\pi^2 b^3}(-\hat{a}_r)$

Relation with Ohm's law (1 mark)

$$Power = I^2 R$$

where

$$R = \frac{\rho \times length}{Area} = \frac{1}{\sigma}\times h \times \frac{1}{\pi b^2}$$

So Power

$$= \frac{I^2 h}{\pi b^2\sigma}$$

which matches with the equation

$$\oint_S \boldsymbol{P}\cdot\boldsymbol{da} = -\sigma\frac{I^2}{\pi^2 b^4\sigma^2}(\pi b^2 h) = \frac{I^2 h}{\pi b^2\sigma}$$

Marking scheme
Total marks = 2 = (1 for Derivation of Poynting vector equation +1 for Relation with Ohm's law)

7. Explain why communication to a satellite has to be done at frequencies larger than 9 MHz, whereas medium wave radio stations operate at frequencies between 0.3 MHz to 3 MHz. Draw a simple sketch showing how the lower frequencies are able to travel from A to B, where A and B are not in line-of-sight of each other. The earths upper atmosphere consists of layers of ionised gases (a plasma) called the ionosphere.

8. A left circularly polarized (LCP) plane wave and a right circularly polarized (RCP) plane wave, with an arbitrary phase difference between them, are co-propagating in the same direction. Prove that the total power carried by the two waves is the sum of powers carried by the individual waves. Can I make such a statement for any two arbitrary polarization states?

Solution: Fields for LCP and RCP with phase ϕ between them,

$$\overrightarrow{E_{lcp}} = E_{10}(\hat{e_x} + j\hat{e_y})e^{j(\omega t-\beta z)} \qquad \overrightarrow{E_{rcp}} = E_{20}e^{j\phi}(\hat{e_x} - j\hat{e_y})e^{j(\omega t-\beta z)}$$
$$\overrightarrow{H_{lcp}} = \frac{E_{10}}{\eta}(-j\hat{e_x} + \hat{e_y})e^{j(\omega t-\beta z)} \qquad \overrightarrow{H_{rcp}} = \frac{E_{20}}{\eta}e^{j\phi}(j\hat{e_x} + \hat{e_y})e^{j(\omega t-\beta z)}$$

Total power flow using the total fields,

$$\begin{aligned}\overrightarrow{P_{total}} &= \frac{1}{2}\,\mathrm{Re}((\overrightarrow{E_{lcp}} + \overrightarrow{E_{rcp}}) \times (\overrightarrow{H_{lcp}} + \overrightarrow{H_{rcp}})^*) \\ &= \frac{1}{2}\,\mathrm{Re}((\overrightarrow{E_{lcp}} \times \overrightarrow{H_{lcp}}^*) + (\overrightarrow{E_{lcp}} \times \overrightarrow{H_{rcp}}^*) + (\overrightarrow{E_{rcp}} \times \overrightarrow{H_{lcp}}^*) + (\overrightarrow{E_{rcp}} \times \overrightarrow{H_{rcp}}^*))\end{aligned}$$

The middle two terms are zero, independent of ϕ. Hence,

$$\overrightarrow{P_{total}} = \frac{1}{2}\,\mathrm{Re}((\overrightarrow{E_{lcp}} \times \overrightarrow{H_{lcp}}^*) + (\overrightarrow{E_{rcp}} \times \overrightarrow{H_{rcp}}^*)) = \overrightarrow{P_{lcp}} + \overrightarrow{P_{rcp}}$$

The statement **<u>cannot</u>** be made for any two arbitrary polarization states, but only for orthogonal states.

Marking scheme
Total marks = 2 = (1 mark for identifying the cross product is 0 and 1 mark for stating it is only true for orthogonal states)

9. You are the manufacturer of sunglasses that have inbuilt polarizers in them. The goal of the sunglasses is to prevent distraction from glare of the sun from smooth surfaces on the road, as may be caused by presence of water or ice on the road. Which one of the following configurations of the polarizer axis for each eye would you choose and why?

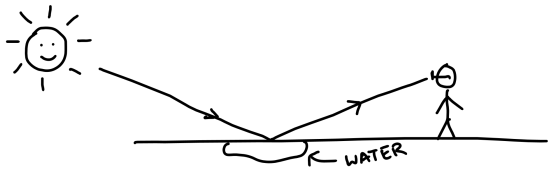


Figure 4: Figure showing glare caused by the reflection from a puddle of water

Note: The two headed arrows in the glasses represent the polarization of light that is allowed to pass through.



(a) (b) (c) (d)

Solution: Option (d). Reflected light around Brewster angle is perpendicularly polarized (TE), i.e. electric field is in the horizontal plane of the eyes, which can be blocked by polarizers with the two headed arrows oriented vertically.

Marking scheme:

Correct option with proper reasoning (mentioning the electric field of the reflected light is in the horizontal plane of the eyes) - **2 marks**

A partial mark is awarded (**1 mark**) if the reasoning is correct but chose the wrong option.

Marks are not awarded if the reasoning is not correct even if the option chosen is correct.

10. Figure 5 represents the E-plane and H-plane radiation patterns of a Hertz dipole antenna, in the far field. Explain how it was plotted (in terms of r, θ, ϕ). What is the difference between the plot in (a) and (b)? What is the orientation of the dipole?

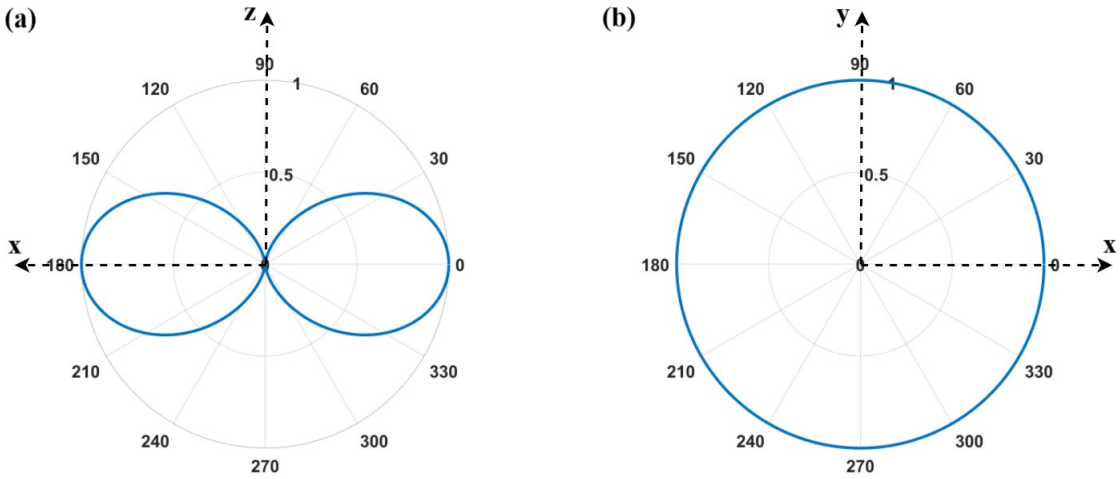


Figure 5

Solution: Plot (a) shows the electric field strength ($|\boldsymbol{E_\theta}|$) as a function of elevation angle (θ). The figure 8 shape represents the $\sin(\theta)$ dependence. Plot (b) shows the magnetic field strength ($|\boldsymbol{H_\phi}|$) as a function of azimuthal angle (ϕ). The cylindrical symmetry of the radiation pattern is exemplified. The relative strength in both plots is deduced by the length of a point on the curve to the origin. The dipole is oriented along the 'z' axis.

Part B This part has 5 questions with 6 marks each.

11. A 100 V/m, 3 GHz plane wave travelling in a perfect dielectric medium with $\epsilon_{r1} = 4$ and $\mu_{r1} = 1$ is incident on a second infinitely thick dielectric slab with $\epsilon_{r2} = 9$ and $\mu_{r2} = 1$. (Assume that the wave is travelling in the + z direction and that the interface lies at z = 0).
    (a) What is the nature of the resultant wave in each region?
    (b) If the second region were not infinitely thick, would your previous answer change? Explain why.
    (c) What is the wavelength of the wave in region 1? Is it the same in region 2?
    (d) Assume the electric field is linearly polarised in the x-direction. Write out an expression for the total field in the first medium, giving values for all relevant terms, like phase constant, etc.
    (e) Calculate the magnetic field $\overrightarrow{H}$ associated with $\overrightarrow{E}$ in the incident medium.

Solution: (a) Travelling wave in the second region and standing wave in the first. This is because there exists incident and reflected waves in the first region.
(b) If the second region was finite, then there may another reflection which would lead to a standing wave in this region as well.
(c) The wavelength in region 1 is obtained from β_1. $\beta_1 = \omega\sqrt{\epsilon_{r1}\mu_{r1}}\sqrt{\epsilon\mu} = 40\pi$ rad/m
$\lambda_1 = 2\pi/\beta_1 = 5cm$. The wavelength will be different in the second medium.
(d) Amplitude is 100 V/m. $\omega = 2\pi\, 3 \times 10^9$ rad/s
Reflection coefficient $\eta_1 = \frac{\eta_0}{\sqrt{\epsilon_{r1}}}$
$\eta_1 = 120\pi/\sqrt{4} = 60\pi$
$\eta_2 = 120\pi/\sqrt{9} = 40\pi$
$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{-20}{100} = -0.2$
The electric field in region 1 is
$\overrightarrow{E_1} = \hat{a_x}100\left(e^{-j\beta_1 z} - 0.2e^{+j\beta_1 z}\right)$ V/m
(e) The magnetic field in region 1 is
$\overrightarrow{H_1} = \hat{a_y}0.53\left(e^{-j\beta_1 z} + 0.2e^{+j\beta_1 z}\right)$ A/m

Marking scheme
Total marks = 6 = (a) 1 mark for stating the standing waves (b) 1 mark for stating the standing waves (c) 1 mark for calculation of wavelength and identifying its different in the second medium, 0 if anyone is missing or just states its different without calculation of wavelength (d) 1 mark for correctly getting the Reflection coefficient with a negative sign. 1 mark to find the E expression correctly, no unit 0 marks (e) 1 mark for correct H expression, no unit 0 marks

12. An air-filled rectangular waveguide has dimensions a = 2 cm and b = 1 cm.
    (a) Determine the range of frequencies over which the guide will operate single mode (TE10).
    (b) What do the indices (10) in the subscript mean?
    (c) Sketch a cross section of the rectangular waveguide showing the electric field lines.
    (d) What does one mean by a mode of a waveguide?
    (e) Why does a rectangular waveguide not support a TEM mode?

Solution:

(a) For TE_{10} mode, $f_c = \frac{c}{2a} = 7.5\ GHz$

For TE_{01} and TE_{20} mode (next order), $f_c = 15\ GHz$

(b) Indices (1,0)

1 - one half wave variation along dimension 'a'.

0 - no variation along dimension 'b'.

(c) Sketch a cross-section of the rectangular waveguide showing the electric field lines.

(d) Distinct E, H distribution/field patterns supported/propagated in a waveguide.

(e) Proof by contradiction - TM fields have to loop - no enclosed currents. Hence contradiction.

Marking scheme

(a) Total 2 marks.

- 1 mark for identifying the lower cut off frequency for single mode operation. i.e, 7.5 GHz
- 1 mark for identifying the upper bound and mentioning the range of frequencies. 7.5 GHz $\leq f_c \leq$ 15 GHz.

(b) Total 1 mark. Need to explain what is m and n.

(c) Total 1 mark. Need to draw Electric field lines looking through the cross section of wave guide.

(d) Total 1 mark. Definition of a mode. Key words mentioned in the solution are verified for full marks

(e) Total 1 mark. Need to prove the non existence of TEM mode by proof of contradiction using Ampere's law and Maxwell's conditions.

13. Consider the circuit shown in Figure 6, where a short-circuited stub is used to maximize the power delivered to the load Z_L= (0.5 - j0.5) Ω. What should be the length (l) of the stub? And, what is the power delivered to the load? Is this the maximum power available from the source? The source frequency and the distributed capacitance of the transmission line are 30 MHz and (100/3) pF/m.

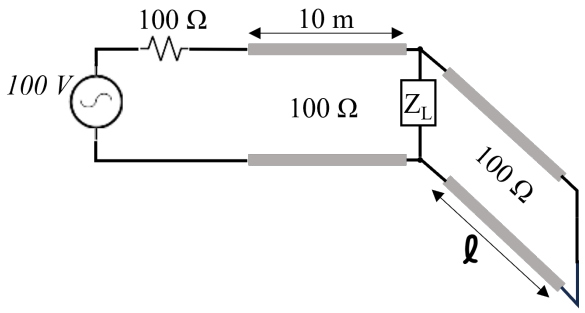


Figure 6: Transmission line circuit for problem 13.

Solution: In order to find l, we need to find β, the propagation constant. We begin by evaluating the distributed inductance of the transmission line:

$$L = CZ_0{}^2 = \frac{100}{3} \times 10^{-12} \times 100^2 \quad H/m$$
$$= \frac{1}{3} \quad \mu H/m$$
$$\beta = \omega\sqrt{LC} = 2\pi \times 30 \times 10^6 \sqrt{\frac{100}{3} \times 10^{-12} \times \frac{1}{3} \times 10^{-6}} = \frac{2\pi}{10} \quad rad.m^{-1} \textbf{(1 mark)}$$

The impedance of the short-circuit stub at the load is given by,

$$Z_{stub} = jZ_0 \tan(\beta l) = j100 \tan(\beta l) \quad \Omega$$

Reactive loads cannot absorb power. Hence, the effective impedance (Z_{eff}) of the combination of the load and the stub should be purely real i.e. $Im(Z_{eff}) = 0$ **(1 mark - (a))**.

$$\frac{1}{Z_{eff}} = \frac{1}{Z_L} + \frac{1}{Z_{stub}} = \frac{1}{0.5 - j0.5} + \frac{1}{j100\tan(\beta l)} \textbf{(1 mark}$$
$$= 1 + j[1 - \frac{1}{100\tan(\beta l)}] \quad \mho$$

Therefore, we need $\tan(\beta l) = 0.01 \implies l = \frac{\tan^{-1}(0.01)}{(\frac{2\pi}{10})}$ i.e. $l \approx \frac{0.1}{2\pi} = 1.592$ cm **(1 mark)**. The wavelength is 10 m, so the impedance seen at the generator is the same as Z_{eff}, which is 1 Ω. The equivalent circuit is thus two resistors 100 Ω and 1 Ω connected in series to the source. Using simple voltage division, the power delivered to the load is

$$P = \frac{1}{2}Re(VI^*) = \frac{1}{2}(\frac{100}{100+1})^2 \approx 0.49 \quad W \textbf{(1 mark - (b))}$$

This is **not** the maximum power available from the source **(1 mark - (c))**. Maximum power transfer occurs when the load impedance is the complex conjugate of the source impedance, in this case 100 Ω.

Marking scheme

(a) Calculation of λ or β :- 1 mark

(b) Calculation of power by RMS values or instantaneous power calculation :- 1 mark.

(c) For maximum power transfer showing either $Z_{th} = Z_s^*$ or $\frac{dP}{dl} = 0$:- 1 mark.

14. Consider two perpendicularly polarized plane waves, of the same frequency ω, incident from either side of the dielectric ($\sigma = 0$ and $\mu_r = 1$) interface as shown in Figure 7. The electric field of the two incident waves are out of phase as in the following expressions.

$$\overrightarrow{E_1} = E_0 e^{-j\mathbf{k_1}.\mathbf{r}} \hat{e_y}$$
$$\overrightarrow{E_2} = -E_0 e^{-j\mathbf{k_2}.\mathbf{r}} \hat{e_y}$$

Find the net electric field in the region $z > 0$ after interaction with the dielectric interface.

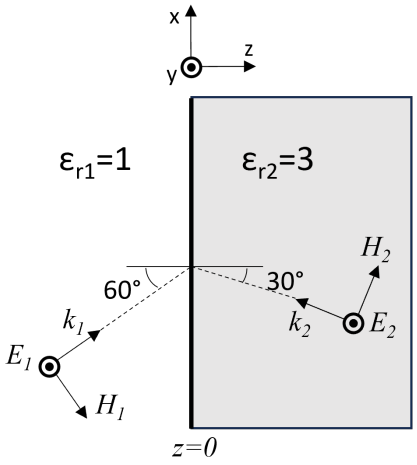


Figure 7: Figure detailing the relative permittivity, approach angles and the coordinate axes for problem 14.

Solution: The reflected and refracted angles can be calculated using Snell's law (phase matching along 'x'). For the wave on the left (wave1), the angle of transmission into the second medium is given by:
$1\sin(60°) = \sqrt{3}\sin(\theta_t) \implies \theta_t = 30°$
For the wave on the right (wave2), the angle of reflection is the same as the angle of incidence, which is also 30°. Hence, the total field is the sum of the incident field of wave2, and the transmitted and reflected fields of wave1 and wave2 respectively. To estimate the magnitudes of these fields we need to evaluate the reflection and transmission coefficients for both waves using Fresnel's equations.
For wave1, the reflection coefficient is

$$\Gamma_1 = \frac{\frac{\eta_2}{\cos(30°)} - \frac{\eta_1}{\cos(60°)}}{\frac{\eta_2}{\cos(30°)} + \frac{\eta_1}{\cos(60°)}} = \frac{\frac{2}{3} - 2}{\frac{2}{3} + 2} = -0.5$$

and so the transmission coefficient $\mathrm{T}_1 = 1 + \Gamma_1 = 0.5$.
For wave2, the reflection coefficient $\Gamma_2 = -\Gamma_1 = 0.5 = \mathrm{T}_1$.
Since the incident fields for wave1 and wave2 are out of phase, the transmitted fields due to wave1 and the reflected fields from wave2 perfectly cancel each other. Therefore, the total field in the region $z > 0$ is the same as that due to wave2.

$$\overrightarrow{E}(z > 0) = -E_0 e^{-j\mathbf{k_2}.\mathbf{r}} \hat{e_y}$$
$$\overrightarrow{H}(z > 0) = -\frac{E_0}{2\eta_2} e^{-j\mathbf{k_2}.\mathbf{r}} (\sqrt{3}\hat{e_x} + \hat{e_z})$$

Marking scheme:

Finding $\theta_t = 30°$ from Snell's law – 1 mark

Finding $\mathrm{T}_1 = 0.5$ – 1 mark

Finding $\Gamma_2 = 0.5 = \mathrm{T}_1$ – 1 mark

Identifying that angle of incidence = angle of reflection for wave 2 and it coincides with transmitted part of wave 1 – 1 mark

Identifying that transmitted fields due to wave1 and the reflected fields from wave2 perfectly cancel each other – 1 mark

Expressing $\overrightarrow{E}(z > 0)$ as $-E_0 e^{-j\mathbf{k_2}\cdot\mathbf{r}}\hat{e_y}$ – 1 mark

15. A left-hand circularly polarized wave impinges at an interface between two different media with an angle of 45°. Determine the state of polarisation of the reflected and transmitted wave if the wave travels from air to a perfect conductor.

Solution: $\theta_i = 45°$
Any arbitrary polarised incident wave can be expressed as $\vec{E_i} = \vec{E_{i||}} + \vec{E_{i\perp}} e^{j\phi}$
Similarly, $\vec{E_r} = \vec{E_{r||}} + \vec{E_{r\perp}} = \Gamma_{||}\vec{E_{i||}} + \Gamma_{\perp}\vec{E_{i\perp}} e^{j\phi}$ and
$\vec{E_t} = \vec{E_{t||}} + \vec{E_{t\perp}} = \tau_{||}\vec{E_{i||}} + \tau_{\perp}\vec{E_{i\perp}} e^{j\phi}$

Given the wave is LCP $\implies |E_{i||}| = |E_{i\perp}|$ and $\phi = \pi/2$
If the wave travels from air to a perfect conductor:
Perfect conductor $\implies \sigma \to \inf \implies \eta_2 = 0$ and hence $\Gamma_{||} = 1$ and $\Gamma_{\perp} = -1$
Thus, $\vec{E_r} = \vec{E_{i||}} - \vec{E_{i\perp}} e^{j\pi/2} = \vec{E_{i||}} + \vec{E_{i\perp}} e^{j\pi} e^{j\pi/2} = \vec{E_{i||}} + \vec{E_{i\perp}} e^{j\pi} e^{j\pi/2} = \vec{E_{i||}} + \vec{E_{i\perp}} e^{-j\pi/2}$

$\therefore$ Reflected wave is right circularly polarised since $|E_{i||}| = |E_{i\perp}|$

There is no transmitted wave in this case as the transmission medium is a perfect conductor.

Marking scheme:

Expressing the wave as a combination of parallel and perpendicular polarisation $\vec{E_i} = \vec{E_{i||}} + \vec{E_{i\perp}} e^{j\phi}$ - **1 mark**

Determining $|E_{i||}| = |E_{i\perp}|$ and $\phi = \pi/2$ for the LCP wave - **1 mark**

Expressing the reflected wave as a combination of parallel and perpendicular polarisation with different reflection coefficients ($\Gamma_{||}$,$\Gamma_{\perp}$) - **1 mark**

Determining the reflection coefficients ($\Gamma_{||}$,$\Gamma_{\perp}$)- **1 mark**

Determining the state of polarisation of the reflected wave as RCP - **1 mark**

Determining that there is no transmitted light - **1 mark**

——————End——————

Useful Formulae

For two vectors $\mathbf{A}(\equiv A_1\hat{e_1} + A_2\hat{e_2} + A_3\hat{e_3})$ and $\mathbf{B}(\equiv B_1\hat{e_1} + B_2\hat{e_2} + B_3\hat{e_3})$, expressed using any orthonormal basis, the scalar or dot product is

$$\boldsymbol{A} \cdot \boldsymbol{B} = A_1B_1 + A_2B_2 + A_3B_3$$

and the vector or cross product is

$$\boldsymbol{A} \times \boldsymbol{B} = \begin{vmatrix} \hat{e_1} & \hat{e_2} & \hat{e_3} \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix}$$

Differential Operator $\boldsymbol{\nabla}$

Cartesian	**Spherical**
Gradient	
$\boldsymbol{\nabla}\psi = \frac{\partial\psi}{\partial x}\hat{e_x} + \frac{\partial\psi}{\partial y}\hat{e_y} + \frac{\partial\psi}{\partial z}\hat{e_z}$	$\boldsymbol{\nabla}\psi = \frac{\partial\psi}{\partial r}\hat{e_r} + \frac{1}{r}\frac{\partial\psi}{\partial\theta}\hat{e_\theta} + \frac{1}{r\sin\theta}\frac{\partial\psi}{\partial\phi}\hat{e_\phi}$
Divergence	
$\boldsymbol{\nabla}\cdot\boldsymbol{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$	$\boldsymbol{\nabla}\cdot\boldsymbol{A} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 A_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin A_\theta) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}A_\phi$
Curl	
$\boldsymbol{\nabla}\times\boldsymbol{A} = \begin{vmatrix} \hat{e_x} & \hat{e_y} & \hat{e_z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$	$\boldsymbol{\nabla}\times\boldsymbol{A} = \frac{1}{r^2\sin\theta}\begin{vmatrix} \hat{e_r} & r\hat{e_\theta} & r\sin\theta\hat{e_\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\phi} \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix}$
Laplacian	
$\boldsymbol{\nabla^2}\psi = \frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2}$	$\boldsymbol{\nabla^2}\psi = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial\psi}{\partial r}) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial\psi}{\partial\theta}) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2}$
$\boldsymbol{\nabla^2 A} = \boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\boldsymbol{A}) - \boldsymbol{\nabla}\times\boldsymbol{\nabla}\times\boldsymbol{A}$	

Constant	**Symbol**	**Value**
Velocity of light	c	3×10^8 (m/s)
Permittivity	ϵ_0	$\frac{1}{36\pi}\times10^{-9}$ (F/m)
Permeability	μ_0	$4\pi\times10^{-7}$ (H/m)
Intrinsic impedance	η_0	120π or 377 (Ω)

Table 1: Constants of free space

Vector identities

$$
\begin{aligned}
(\boldsymbol{A}\times\boldsymbol{B})\cdot\boldsymbol{C} &= (\boldsymbol{B}\times\boldsymbol{C})\cdot\boldsymbol{A} = (\boldsymbol{C}\times\boldsymbol{A})\cdot\boldsymbol{B} \\
(\boldsymbol{A}\times\boldsymbol{B})\times\boldsymbol{C} &= (\boldsymbol{A}\cdot\boldsymbol{C})\boldsymbol{B} - (\boldsymbol{A}\cdot\boldsymbol{B})\boldsymbol{C} \\
\boldsymbol{\nabla}\cdot(V\boldsymbol{A}) &= \boldsymbol{A}\cdot\boldsymbol{\nabla}V + V\boldsymbol{\nabla}\cdot\boldsymbol{A} \\
\boldsymbol{\nabla}\times(V\boldsymbol{A}) &= \boldsymbol{\nabla}V\times\boldsymbol{A} + V\boldsymbol{\nabla}\times\boldsymbol{A} \\
\boldsymbol{\nabla}\cdot(\boldsymbol{A}\times\boldsymbol{B}) &= \boldsymbol{B}\cdot\boldsymbol{\nabla}\times\boldsymbol{A} - \boldsymbol{A}\cdot\boldsymbol{\nabla}\times\boldsymbol{B} \\
\boldsymbol{\nabla}(\boldsymbol{A}\cdot\boldsymbol{B}) &= (\boldsymbol{A}\cdot\boldsymbol{\nabla})\boldsymbol{B} + (\boldsymbol{B}\cdot\boldsymbol{\nabla})\boldsymbol{A} + \boldsymbol{A}\times(\boldsymbol{\nabla}\times\boldsymbol{B}) + \boldsymbol{B}\times(\boldsymbol{\nabla}\times\boldsymbol{A}) \\
\boldsymbol{\nabla}\cdot\boldsymbol{\nabla} &\equiv \boldsymbol{\nabla^2} \\
\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A}) &= 0 \\
\boldsymbol{\nabla}\times(\boldsymbol{\nabla}V) &= 0
\end{aligned}
$$